

QUARTUS II Quick Reference Guide

By Professor Cesar Ortega-Sanchez and Azadeh Nazemi

The purpose of this guide is to provide some basic pointers so that students can start using the QUARTUS II development software. There is no better way of learning a tool than by using it; hence, students are encouraged to discover the tools by themselves.

Introduction

QUARTUS II is a development environment created by Altera to support its Field-Programmable Logic Devices. A typical design flow using QUARTUS II is shown in Fig.1.

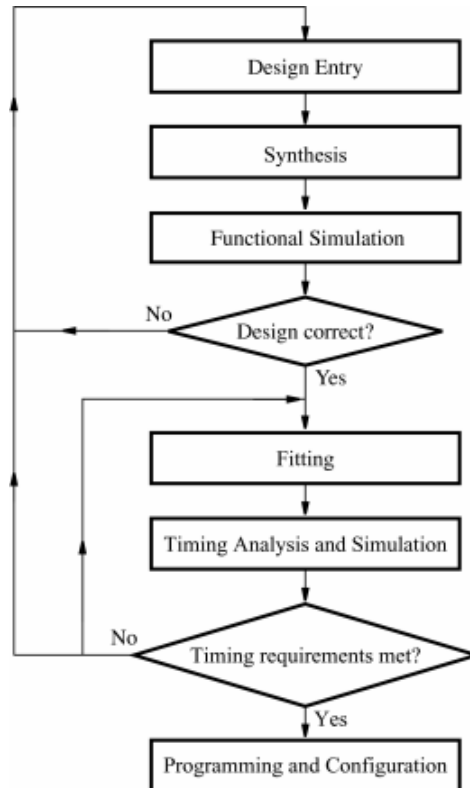


Figure 1. QUARTUS II design flow

Designs can be entered using schematic diagrams, state machines, hardware description languages (HDL) or any combination of them.

Once the design has been entered, it is verified by simulation. Simulation helps the designer to observe the expected behaviour of circuits regardless of the device that is going to be used. If errors are found, the design is corrected and new simulations performed. This cycle is repeated until a satisfactory design is obtained.

Verified designs are synthesized for the particular device that is going to be used (e.g. FPGA, CPLD). Synthesised designs are simulated taking into account particular features of each device (delays, embedded devices, etc.). If functional and timing simulations are not satisfactory, then the design is modified and the whole cycle repeated.

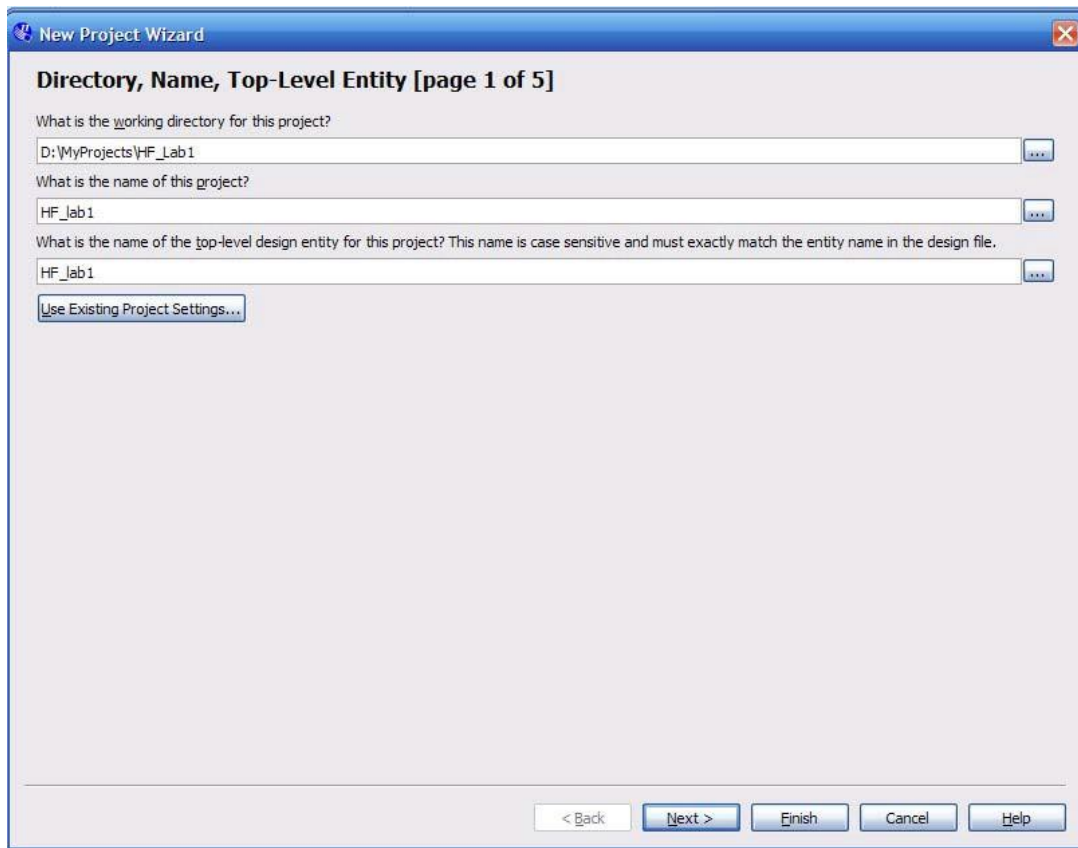
Successfully Synthesised designs are translated into configuration files that are downloaded to the selected programmable logic device. For the purpose of this laboratory, a prototyping board with an Altera Cyclone III FPGA will be used.

The following sections present only the basic steps that have to be followed to create, simulate and test a digital design. More advanced features have to be investigated and/or discovered by the students.

Create a new project using schematics as main sources

To create a new project:

1. Find the Quartus II icon in your computer and launch the program.
2. In the main screen select **File → New Project Wizard...** New Project Wizard appears.
3. Enter or browse to a location (directory path with no spaces) for the new project.
4. Type a meaningful name in the **Project Name** field and click **Next**.



The screenshot shows the 'New Project Wizard' dialog box, titled 'New Project Wizard'. The main area is labeled 'Directory, Name, Top-Level Entity [page 1 of 5]'. It contains three text input fields with browse buttons (three dots) to the right:

- First field: 'What is the working directory for this project?' with the text 'D:\MyProjects\HF_Lab1'.
- Second field: 'What is the name of this project?' with the text 'HF_Lab1'.
- Third field: 'What is the name of the top-level design entity for this project? This name is case sensitive and must exactly match the entity name in the design file.' with the text 'HF_Lab1'.

Below these fields is a button labeled 'Use Existing Project Settings...'. At the bottom of the dialog are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

- In the following panel select existing files you want to add to your new project. If you want to create a brand new file click **Next**.

Add Files [page 2 of 5]

Select the design files you want to include in the project. Click Add All to add all design files in the project directory to the project.
Note: you can always add design files to the project later.

File name:

File Name	Type	Library	Design Entry/Synthesis Tool	HDL Version
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Specify the path names of any non-default libraries.

< Back Next > Finish Cancel Help

- Select **Family= Cyclone III**, **Device Name= EP3C16F484C6**. Click **Next**.

Family & Device Settings [page 3 of 5]

Select the family and device you want to target for compilation.

Device family
Family:
Devices:

Target device
☐ Auto device selected by the Fitter
☒ Specific device selected in 'Available devices' list
☐ Other: n/a

Show in 'Available devices' list
 Package:
 Pin count:
 Speed grade:
 Name filter:
☒ Show advanced devices ☐ HardCopy compatible only

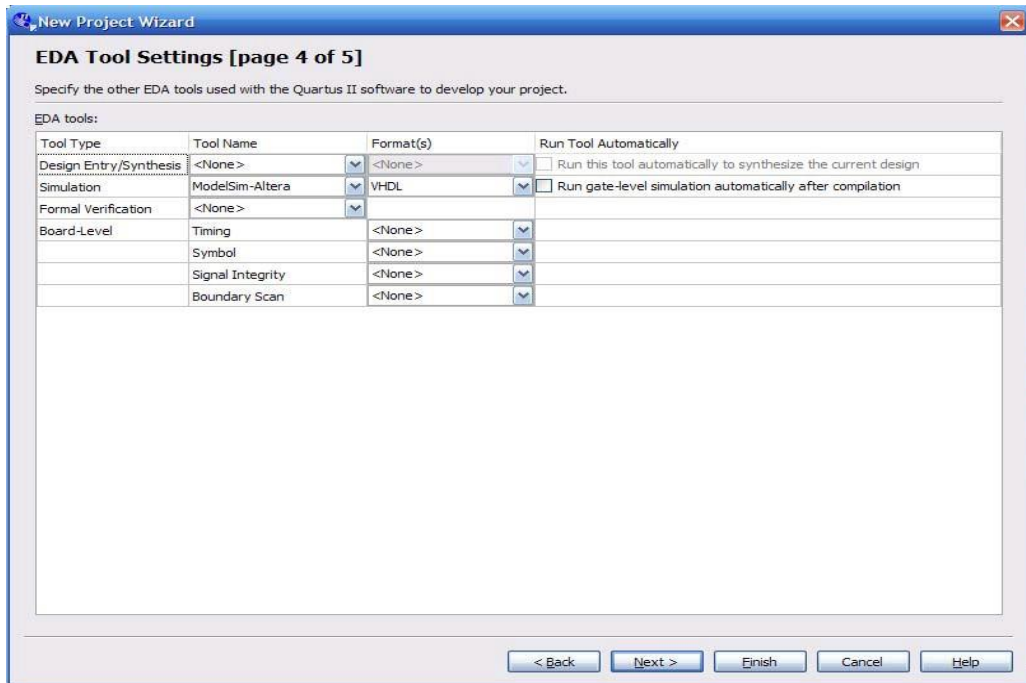
Available devices:

Name	Core Voltage	LEs	User I/Os	Memory Bits	Embedded multiplier 9-bit elements	PLL	SLC
EP3C16F256C7	1.2V	15408	169	516096	112	4	20
EP3C16F256C8	1.2V	15408	169	516096	112	4	20
EP3C16F256I7	1.2V	15408	169	516096	112	4	20
EP3C16F484C6	1.2V	15408	347	516096	112	4	20
EP3C16F484C7	1.2V	15408	347	516096	112	4	20
EP3C16F484C8	1.2V	15408	347	516096	112	4	20
EP3C16F484I7	1.2V	15408	347	516096	112	4	20
EP3C16M164...	1.2V	15408	93	516096	112	4	20

Companion device
 HardCopy:
☐ Limit DSP & RAM to HardCopy device resources

< Back Next > Finish Cancel Help

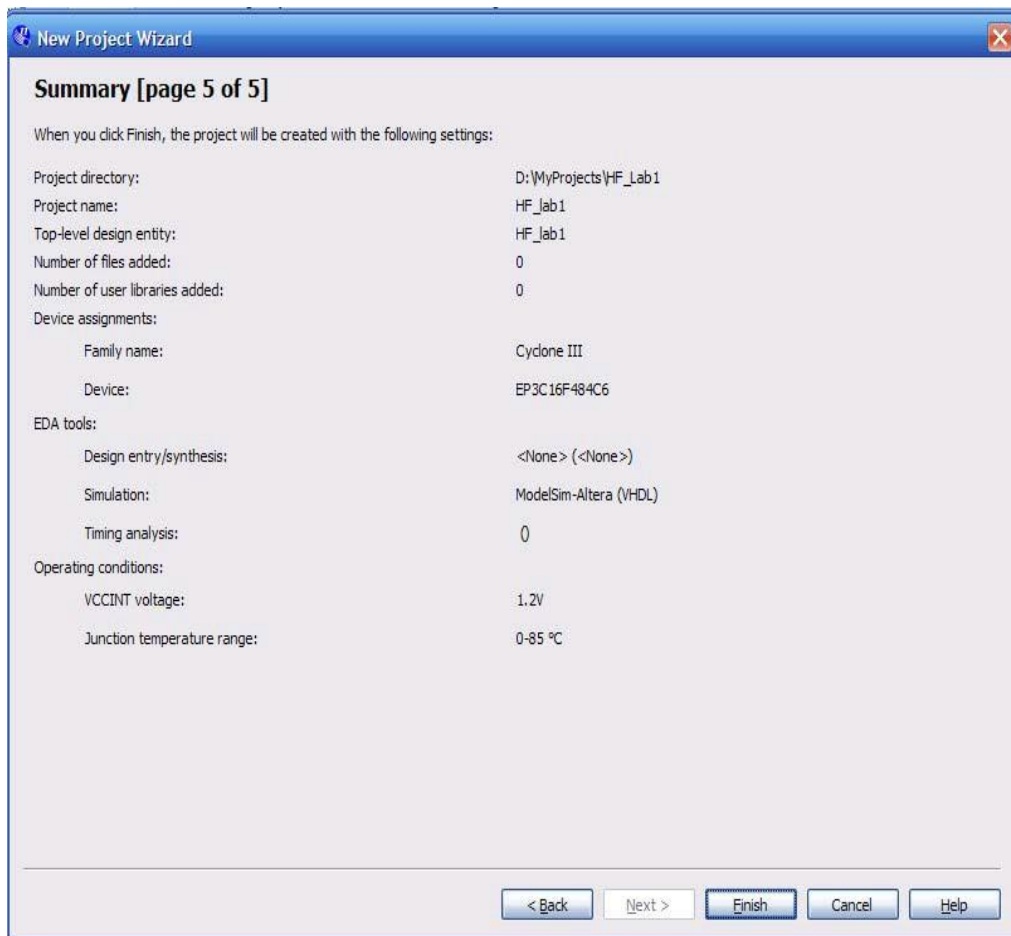
7. Leave the default values in the remaining fields. Click **Next**.



The screenshot shows the 'EDA Tool Settings' window, page 4 of 5. It contains a table for configuring EDA tools. The 'Design Entry/Synthesis' tool is set to '<None>' with format '<None>'. The 'Simulation' tool is set to 'ModelSim-Altera' with format 'VHDL'. The 'Formal Verification' tool is set to '<None>'. The 'Board-Level' section includes 'Timing' (set to '<None>'), 'Symbol' (set to '<None>'), 'Signal Integrity' (set to '<None>'), and 'Boundary Scan' (set to '<None>'). There are checkboxes for 'Run Tool Automatically' for synthesis and simulation, both of which are unchecked. Navigation buttons at the bottom include '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Tool Type	Tool Name	Format(s)	Run Tool Automatically
Design Entry/Synthesis	<None>	<None>	☐ Run this tool automatically to synthesize the current design
Simulation	ModelSim-Altera	VHDL	☐ Run gate-level simulation automatically after compilation
Formal Verification	<None>		
Board-Level	Timing	<None>	
	Symbol	<None>	
	Signal Integrity	<None>	
	Boundary Scan	<None>	

8. Make sure the summary at the end of the process looks like this. Click **Finish**.



The screenshot shows the 'Summary' window, page 5 of 5. It displays the project settings that will be used when the project is created. The settings include: Project directory (D:\MyProjects\HF_Lab1), Project name (HF_lab1), Top-level design entity (HF_lab1), Number of files added (0), Number of user libraries added (0), Device assignments (Family name: Cyclone III, Device: EP3C16F484C6), EDA tools (Design entry/synthesis: <None> (<None>), Simulation: ModelSim-Altera (VHDL), Timing analysis: 0), and Operating conditions (VCCINT voltage: 1.2V, Junction temperature range: 0-85 °C). Navigation buttons at the bottom include '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

When you click Finish, the project will be created with the following settings:

Project directory: D:\MyProjects\HF_Lab1

Project name: HF_lab1

Top-level design entity: HF_lab1

Number of files added: 0

Number of user libraries added: 0

Device assignments:

- Family name: Cyclone III
- Device: EP3C16F484C6

EDA tools:

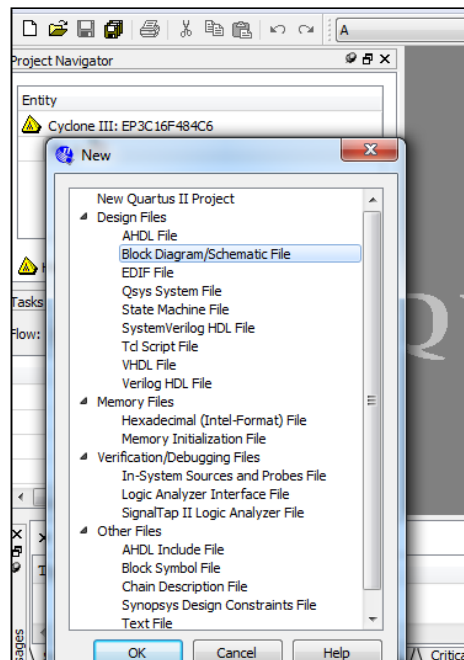
- Design entry/synthesis: <None> (<None>)
- Simulation: ModelSim-Altera (VHDL)
- Timing analysis: 0

Operating conditions:

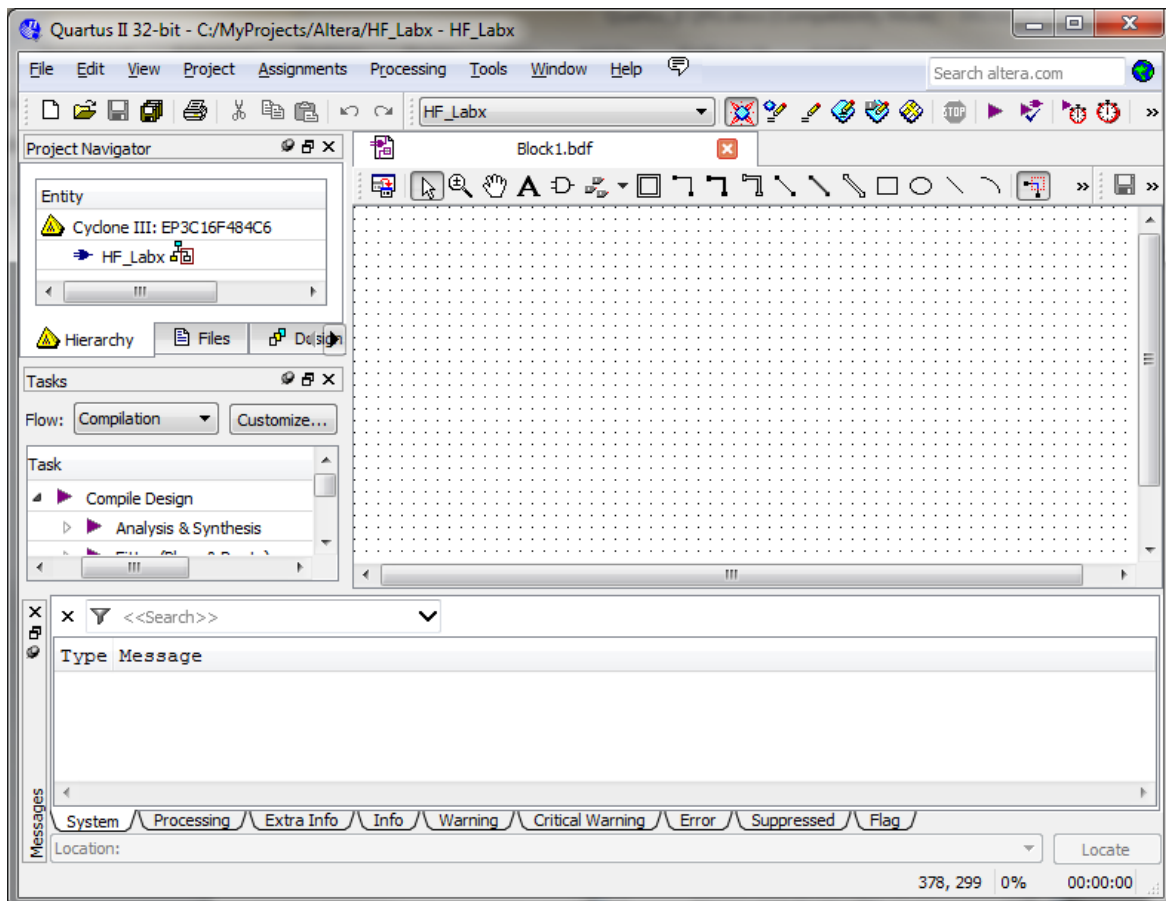
- VCCINT voltage: 1.2V
- Junction temperature range: 0-85 °C

Creating a schematic source file for the project

To create a schematic select **File→New→** From the list choose: **Block Diagram/Schematic File** and click **OK**.

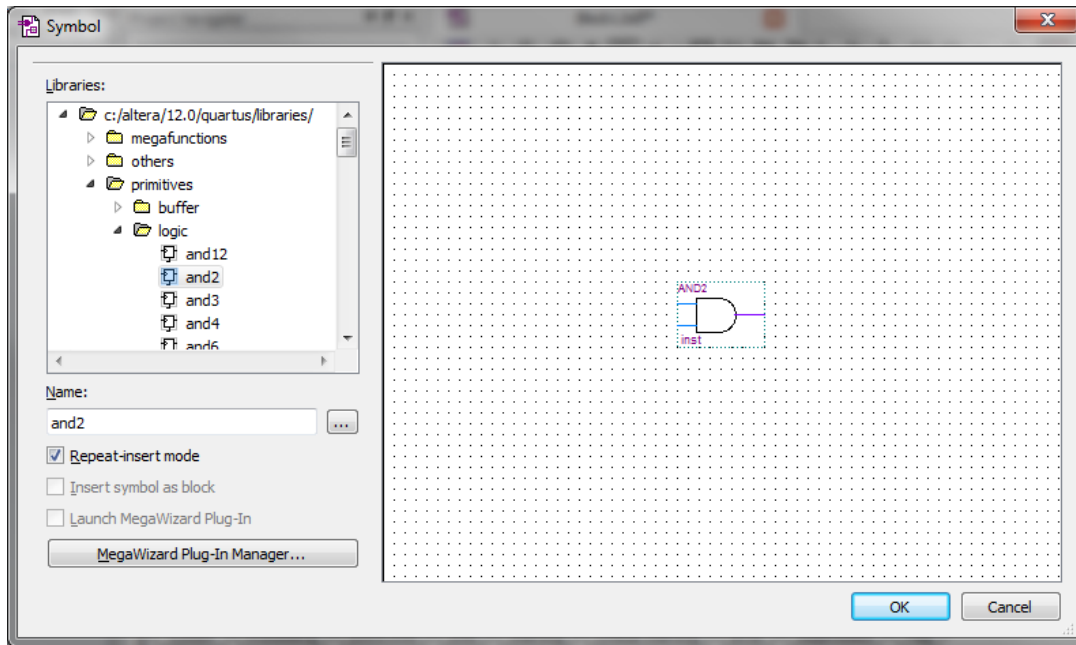


You will be presented with a clean area where you will create your new schematic.



Adding Logic-Gate Symbols

The Graphic Editor provides a number of libraries which include circuit elements that can be imported into a schematic. Double-click on the blank space in the Graphic Editor window, or click on the icon in the toolbar that looks like an AND gate . The following pop-up box will appear.

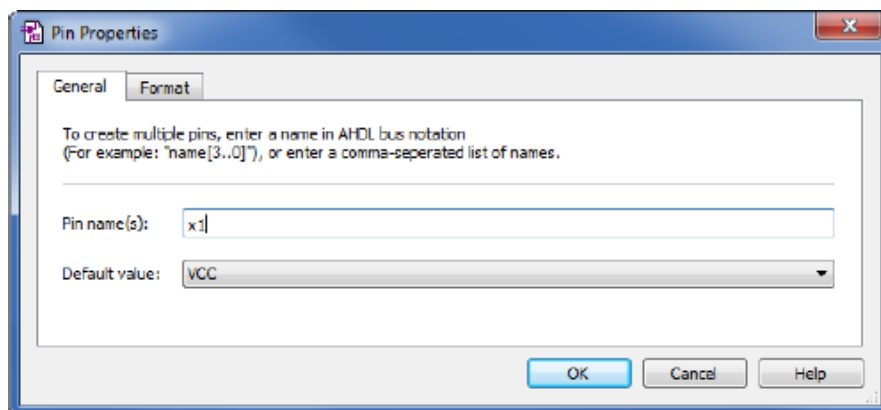


To add a **two-input AND gate** first you have to find it in the **Libraries** box hierarchy. Expand **libraries -> primitives -> logic** and select **and2**, which is a two-input AND gate, and click OK. Now, the AND gate symbol appears in the Graphic Editor window. Using the mouse, move the symbol to a desired location and click to release it there.

Adding Symbols for Inputs and Outputs

Having entered the logic-gate symbols, it is now necessary to enter the symbols that represent input and output ports of the circuit. Use the same procedure as for adding gates, but choose the port symbols from the **library -> primitives -> pin** or use the icon .

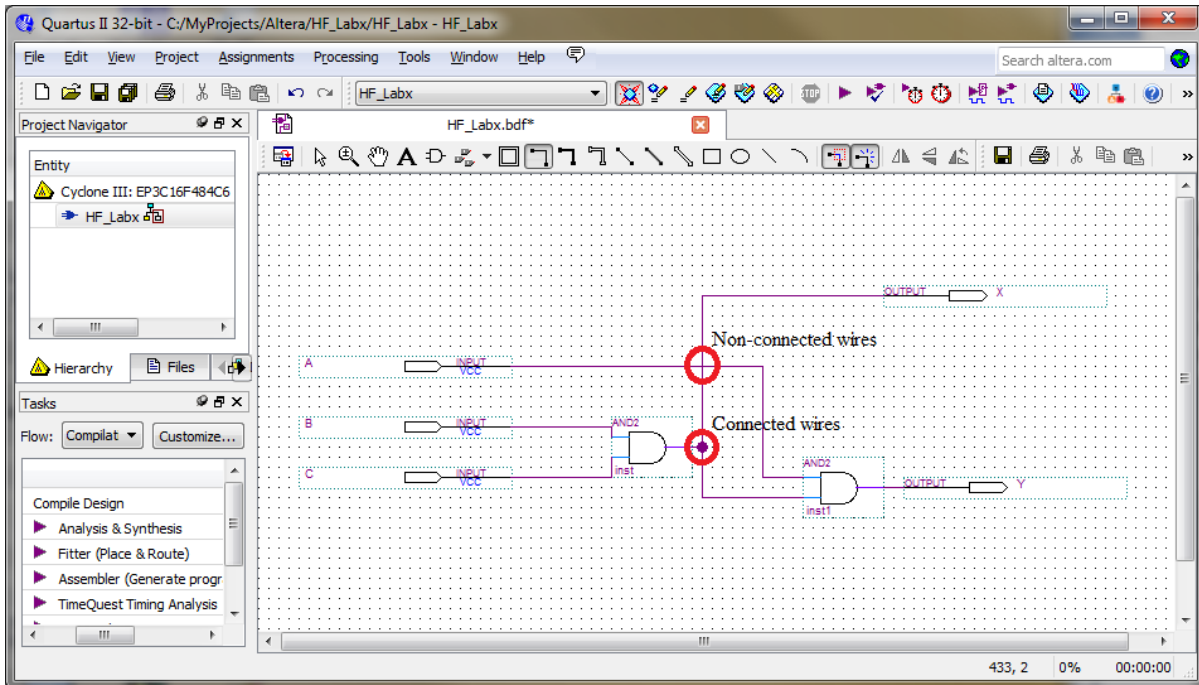
To assign names to input and output symbols double-click on them and type the pin's name in the dialog box that appears.



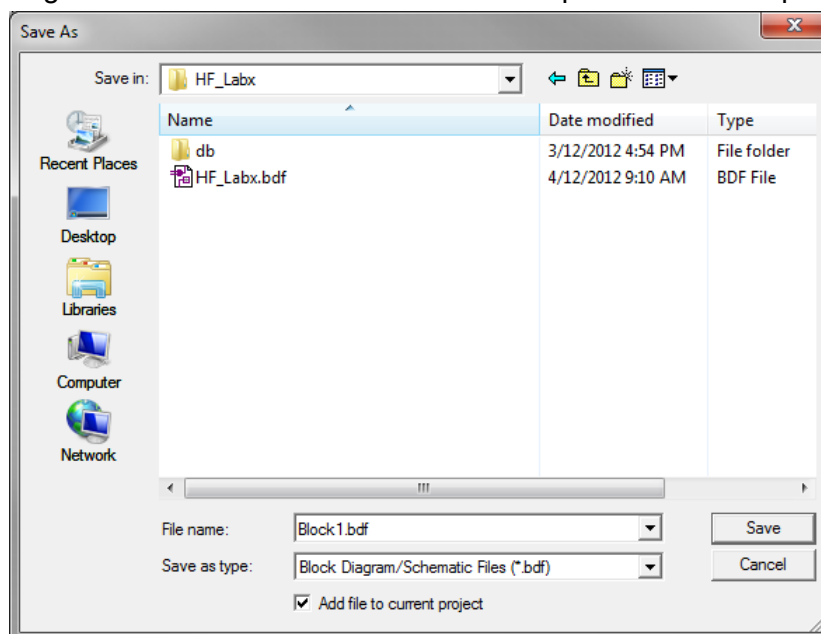
Notice that these inputs and outputs are not yet connected to any FPGA pins. That step will come later (see Pin Assignment).

Connecting Nodes with Wires

The symbols in the diagram have to be connected by drawing lines (wires). Click on the icon  in the toolbar to activate the **Orthogonal Node Tool**. To add a wire position the mouse on an input, output or wire and press the left mouse button. While keeping the left button pressed drag the mouse to the end point of the line you are drawing. Release the button when the mouse is on the wire's end point desired position. Make sure wires touch the actual connection points of gates, inputs and outputs. Notice that when two wires are connected a dot appears between them.



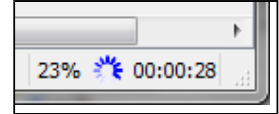
Once you have completed your schematic diagram, save it using the icon . Make sure you save the file in a folder with the name of your project. If it does not exist, create it. Give your files meaningful names. Several schematics can be part of the same project.



Compiling your Design

The entered schematic diagram file (.bdf), is processed by several Quartus II tools that analyse the file, synthesise the circuit, and generate an implementation of it for the target chip. These tools are controlled by the application program called the Compiler.

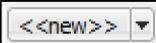
Run the Compiler by selecting **Processing → Start Compilation**, or by clicking on the icon . Your project must be saved before compiling. As the compilation moves through various stages, its progress is reported in a panel on the left side of the Quartus II interface. Also, a progress icon is shown in the bottom, right-hand side corner of the interface window:

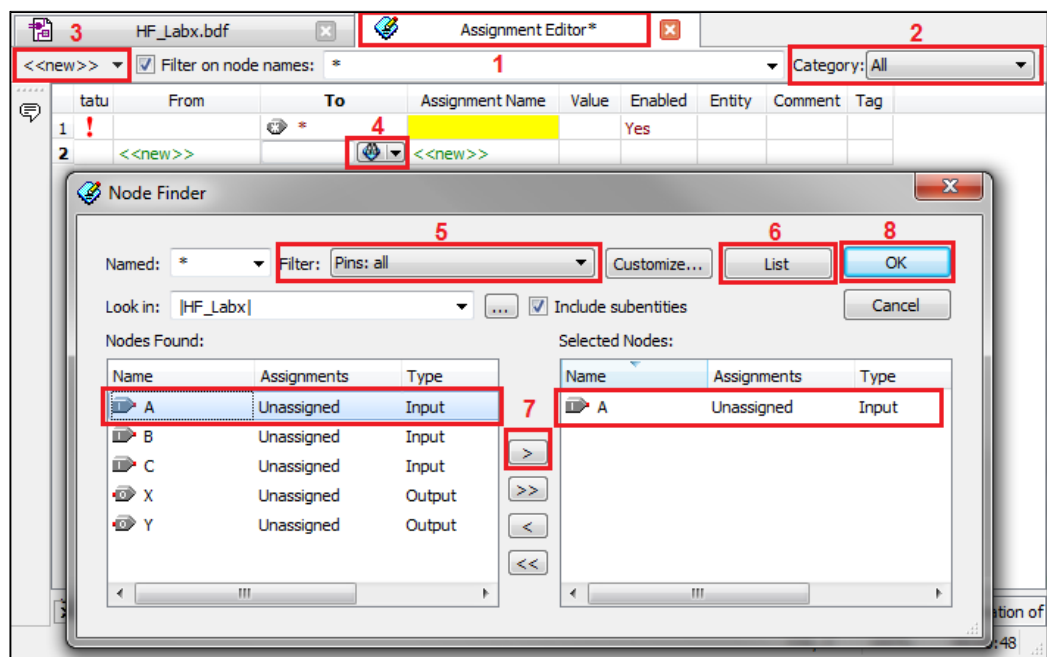


Successful (or unsuccessful) compilation is indicated in a pop-up box. Various messages are displayed in the message window. In case of errors, appropriate messages will be given. When the compilation is finished, a compilation report is produced; it can be opened at any time by selecting **Processing → Compilation Report** or by clicking on the icon .

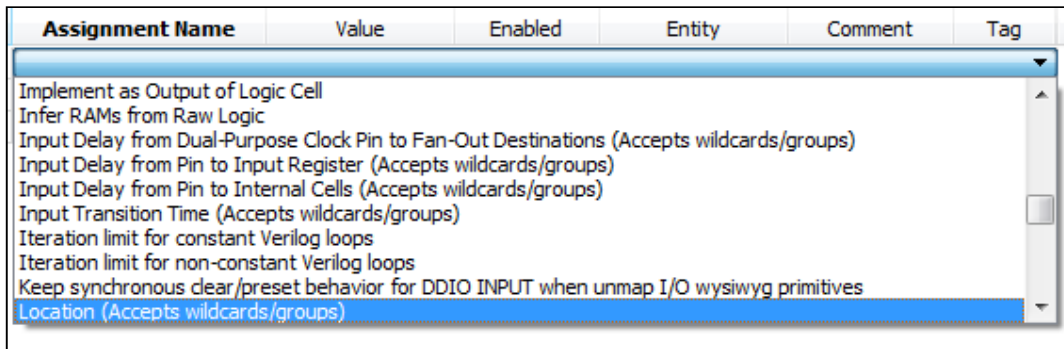
Pin Assignment

Pin assignments are made by doing the following steps:

- 1) Select **Assignments → Assignment Editor** in the main menu. A new panel appears. Numbers in the following figure correspond with these steps.
- 2) In the **Category** drop-down menu select **All**.
- 3) Click on the  button located near the top left corner to make a new item appear in the table.
- 4) Double click the box under the column labelled **To** so that the Node Finder button  appears. Click on the button (not the drop down arrow).
- 5) In the Filter drop-down menu select **Pins: all**.
- 6) Then click the **List** button to display the input and output pins to be assigned.
- 7) Click on the first pin to be assigned and click the **>** button; this will enter the pin in the **Selected Nodes** box. The **>>** button selects all inputs and outputs.
- 8) Click OK. The newly added pin(s) should appear in the column labeled **To**.

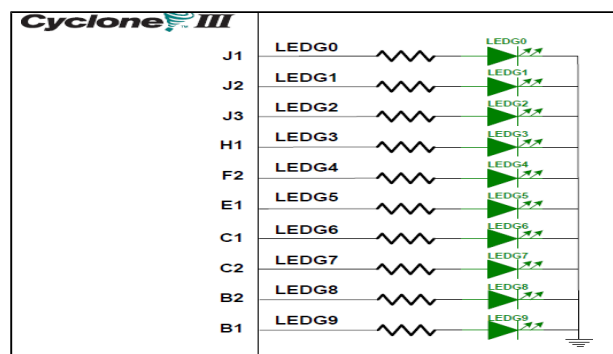
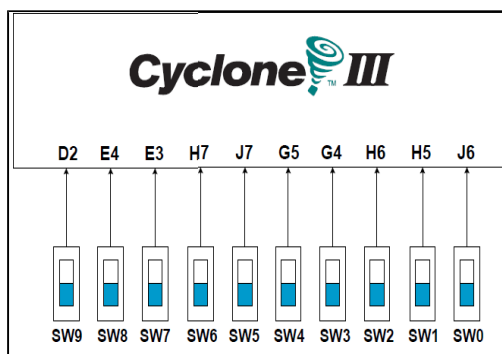


- 9) For every input and output double-click on the empty box in the column labelled **Assignment Name**. A drop-down menu appears. Scroll down and select **Location (Accepts wildcards/groups)**. Type 'L' to quickly find it.



- 10) For every input and output fill the **Value** column with the names of the FPGA pins that you want connected to your inputs and outputs. Some of these pins are externally connected to the DE0 board resources, like LEDs and switches; others are connected to external connectors for you to use in any way you want. The location of these resources can be found in the **DE0 board user manual** available from Blackboard. The names of pins in the **Value** column must begin with **PIN_** followed by the identifier shown in the user manual. In the following example input **x1** in the schematic diagram is being connected to pin **J6**, and pin J6 is externally connected to a switch on the board (**SW0**). Also, output **y1** in the schematic diagram is being connected to pin **J1**, and consequently to an LED (**LEDG0**).

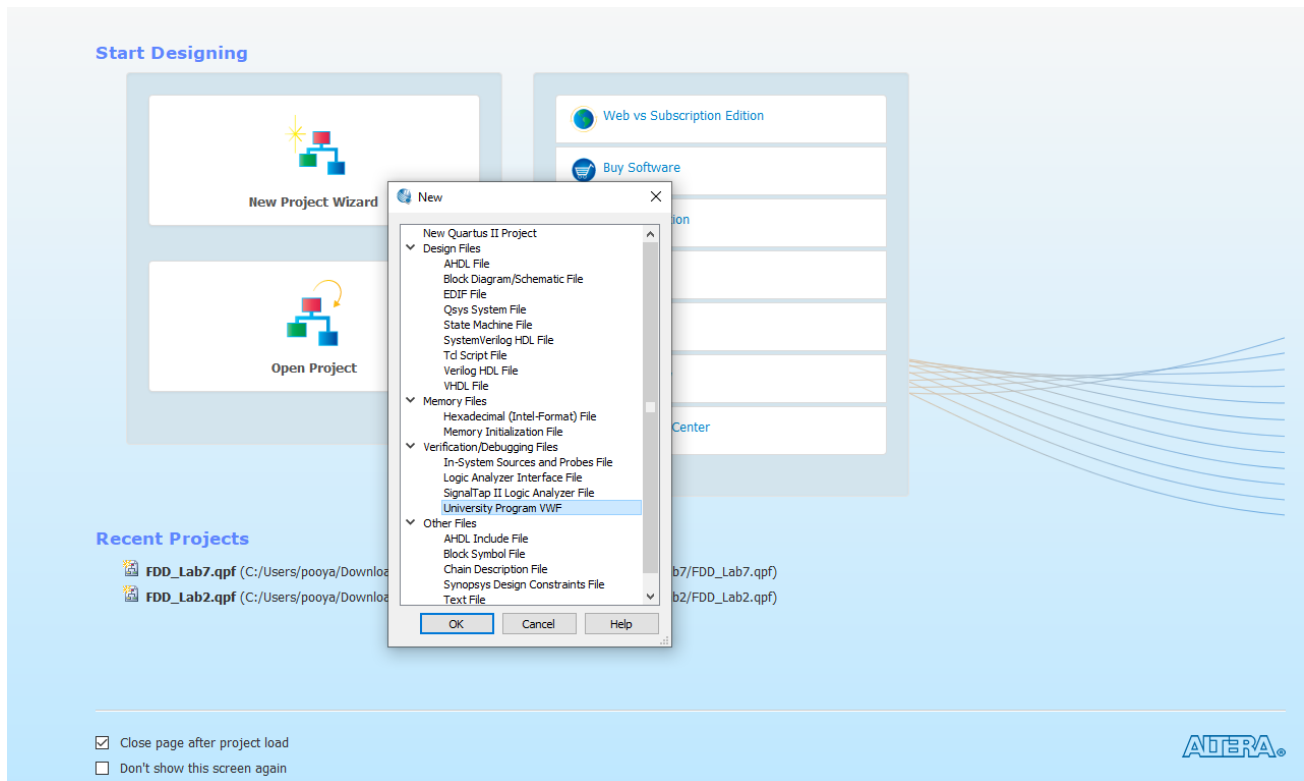
<<new>>		Filter on node names: *		Category:				
tatu	From	To	Assignment Name	Value	Enabled	Entity	Comment	Tag
1	✓	y1	Location	PIN_J1	Yes			
2	✓	x1	Location	PIN_J6	Yes			
3	<<new>>	<<new>>	<<new>>					



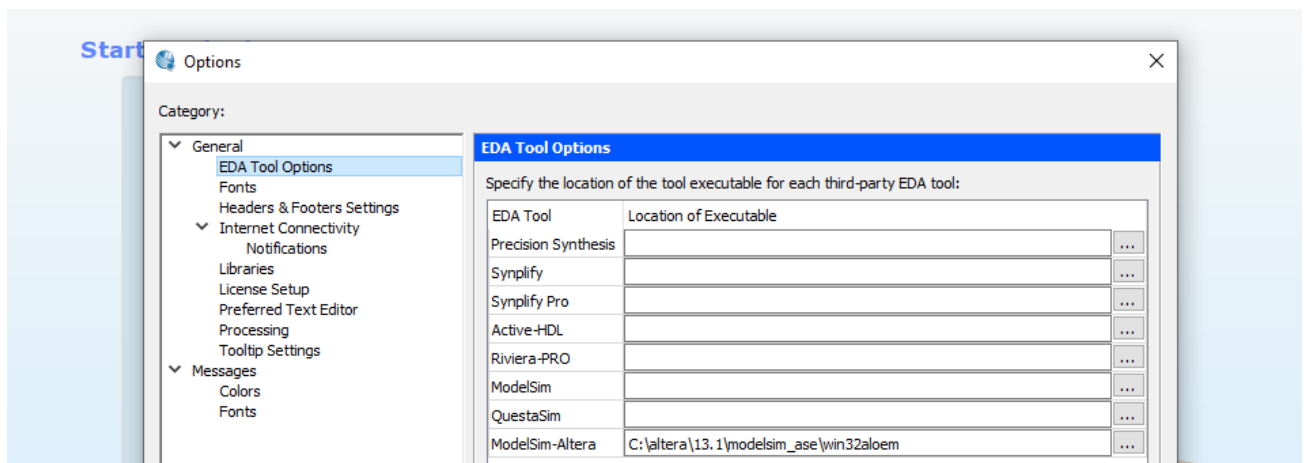
- 11) Once you have finished allocating all you inputs and outputs save your work. A file with extension **QSF** (Quartus II Settings File) will be generated. Open this file with an external text editor and observe the format in which information is stored.

Simulating your design

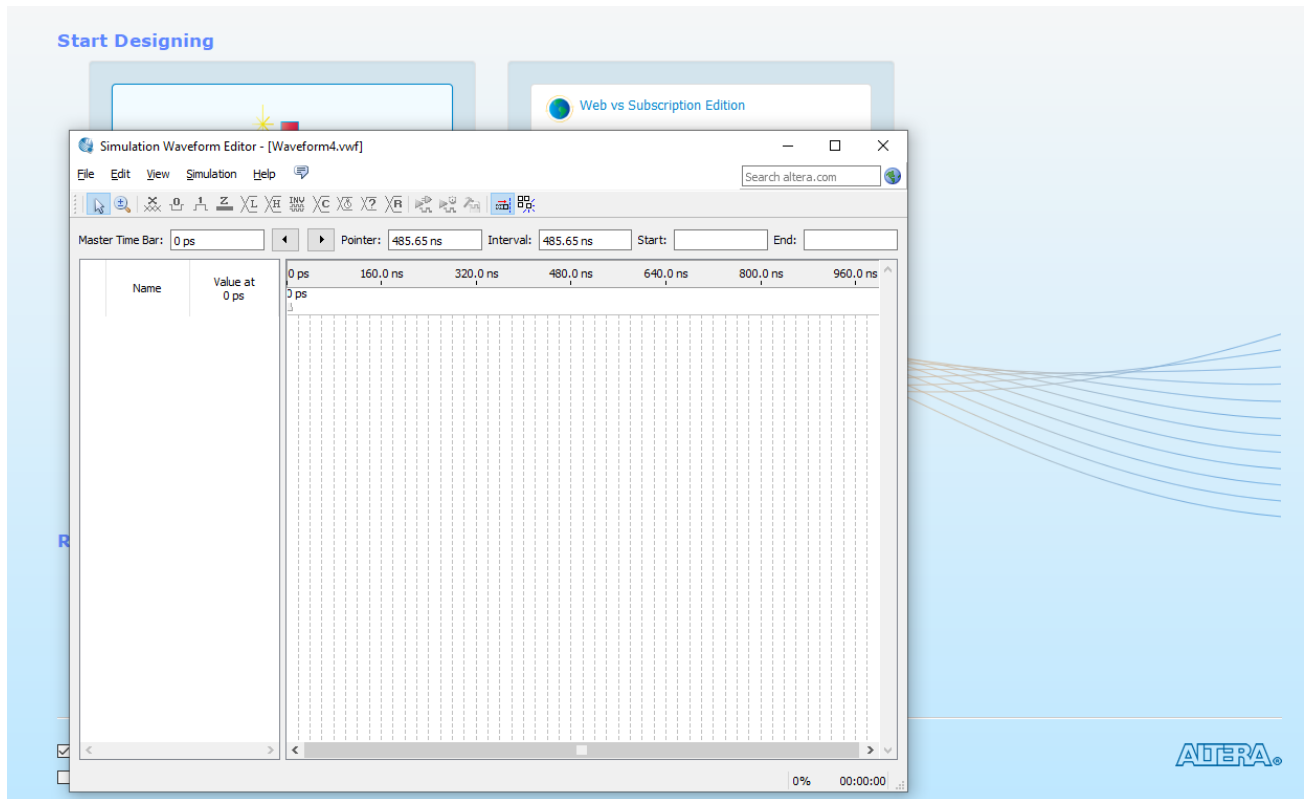
To verify your design, you need to simulate it. Simulation means that you will stimulate your design with a set of inputs and observe the outputs to verify that your circuit produces the expected behaviour. There are several simulators available in Altera; in this unit we will be using **ModelSim**. ModelSim is a tool integrated into the Quartus Altera environment; hence you can launch it directly from within Quartus. In Quartus II, go to **File** → **New**, then select **University Program VWF** from the Verification/Debugging Files section to open a new simulation waveform editor.



Note: if you encounter errors, verify the ModelSim path first: go to Tools → Options → EDA Tool Options (under General). Ensure the ModelSim-Altera entry points to a directory similar to **C:\...\\altera\\13.1\\modelsim_ase\\win32aloem** (adjust based on your installation).

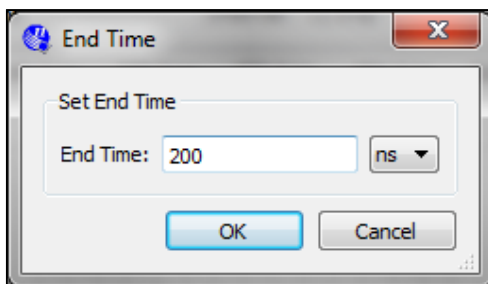


The following screenshot shows a view of ModelSim main window.

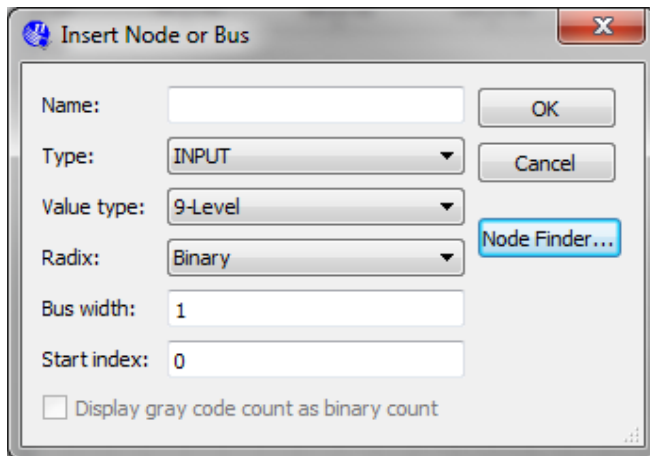


To generate an input file (called **test bench**) and simulate your design, follow these steps:

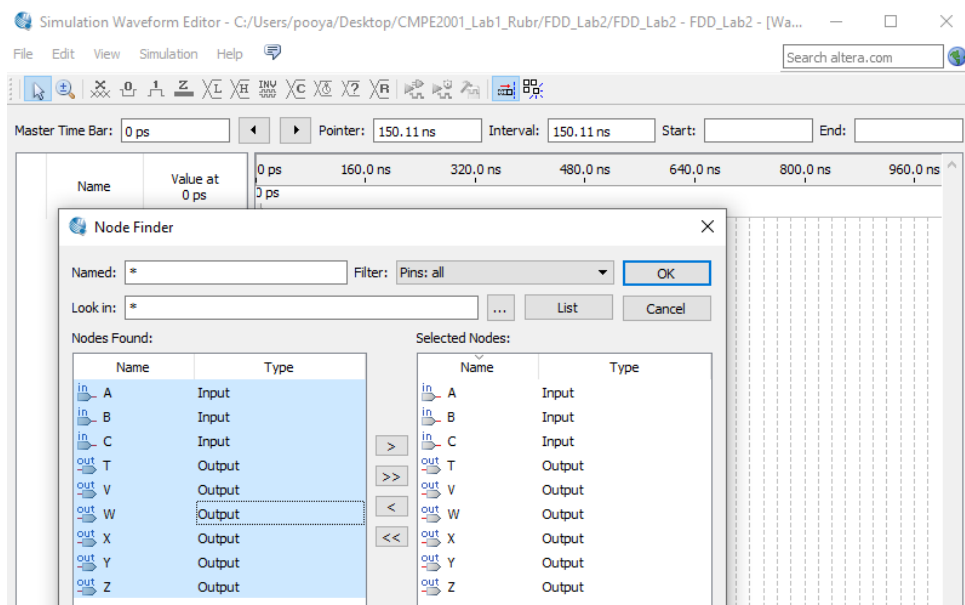
1. Select **File** → **Open Project** and select your Quartus II Project File (**.qpf file**) located in the folder where your schematic diagram was saved.
2. Open the Waveform Editor window (ModelSim) by selecting **File** → **New**, then select **University Program VWF**.
3. Set the desired simulation time. To simulate from 0 to 200 ns select **Edit** → **Set End Time** and enter 200 ns in the dialog box that pops up.



4. Include the input and output nodes of the circuit to be simulated. Select **Edit** (in simulation waveform editor) → **Insert** → **Insert Node or Bus** → **Node Finder**

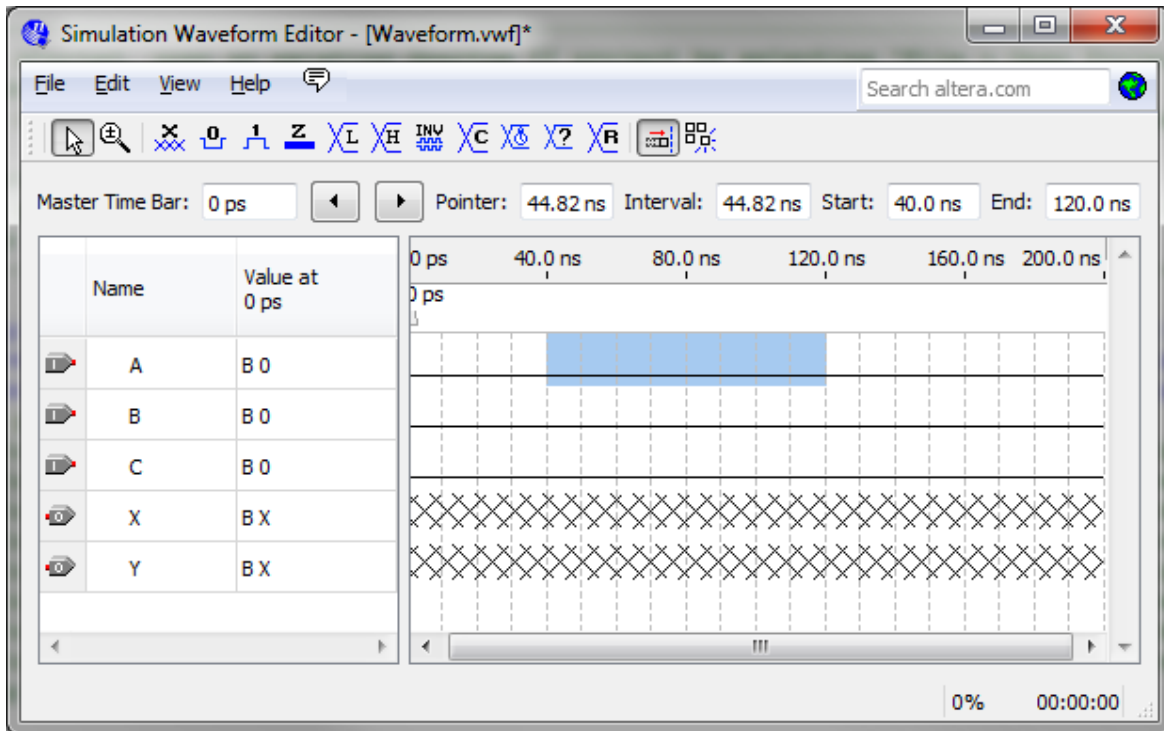


Then select **List** and copy the required Nodes from **Nodes Found** to **Selected Nodes** using the **Arrows**.

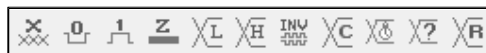


Note: If the Nodes Found list is empty when you click List, it's usually because the design hasn't been compiled yet. Go back to the main Quartus window, select Processing → Start Compilation (or press the purple triangle icon), wait for it to finish successfully, then re-open the Node Finder or restart the simulator after compilation.

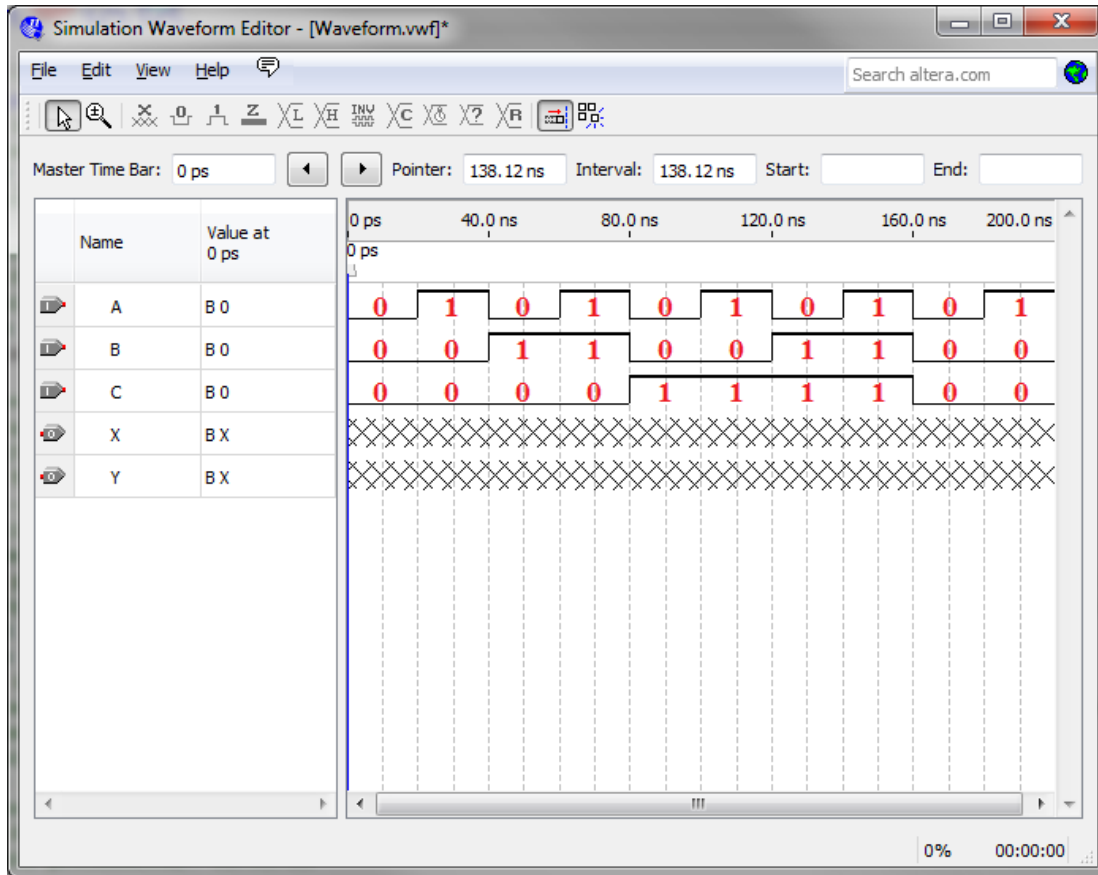
Your inputs and outputs now appear in the Simulation Waveform Editor. Notice that all inputs have a default value of zero and all outputs are undefined.



- To change the value of your inputs, select the time interval you are interested in by pressing the left button of your mouse at the start of the interval and dragging it to the end time. The interval of interest should be highlighted when you release the button. In the previous figure the interval 40 to 120 ns have been selected for input **A**.
- Use the icons in the tool bar to select a value for the highlighted interval. Try all the icons to see what effect they produce on the input signals. Some of these icons only apply to buses.

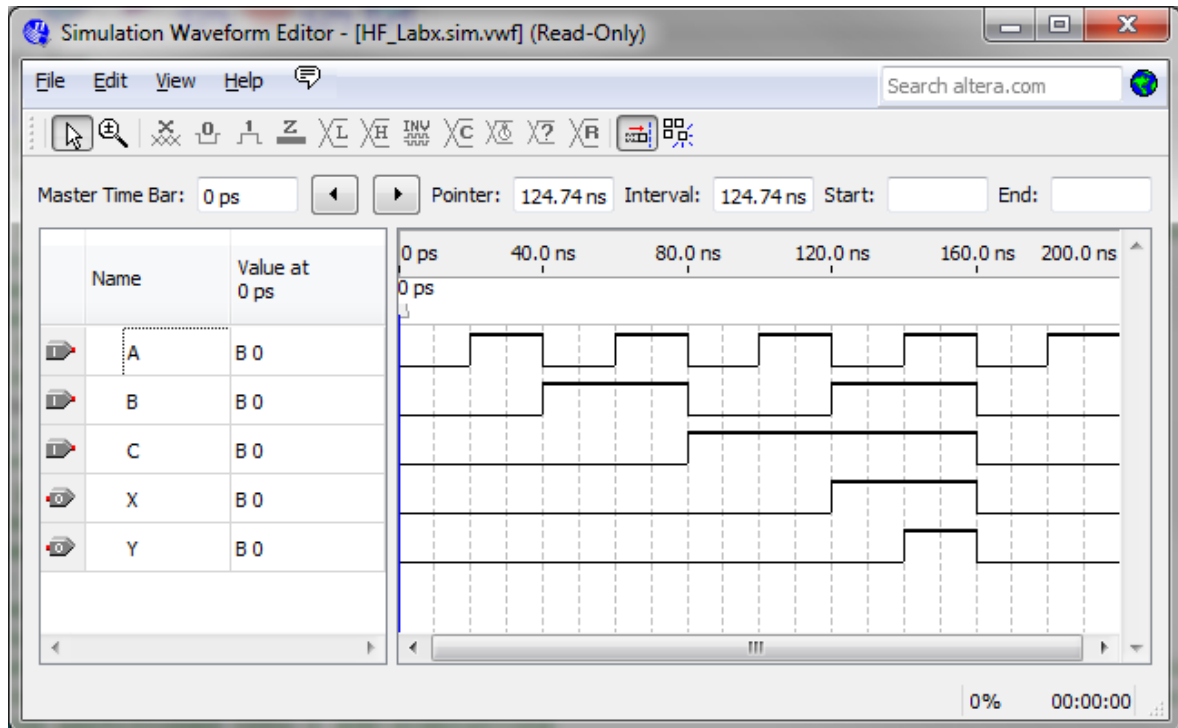


You can use the **counter** icon  to quickly create all possible combinations of inputs for combinational circuits. This method has been used in the following figure to set inputs to all possible combinations of three binary variables, from 000 to 111. Play with the waveform editor until you can comfortably reproduce what is shown in the figure (for clarity, the **red numbers** were added to the figure).



7. Save the file using **File -> Save**. This will create a file called **Waveform.vwf** in your project's folder. If you are planning to use different test benches in the same project, then use **File -> Save As** and save each test bench with a different name.
8. To perform the functional simulation, return to the Simulation Waveform Editor Window and select **Simulation -> Run Functional Simulation**.
9. If the simulation runs successfully, a 'Simulation Flow Progress' window will pop up and show each step completing. It will then close, and a new window will open with the simulation results (your input waveforms plus the computed outputs).

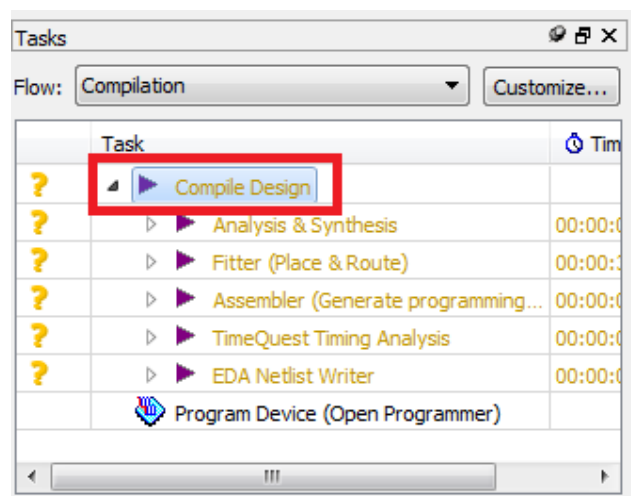
Finally, a new window opens showing the results of your simulation. This view is very similar to the waveform editor, except that outputs will show values corresponding to every combination of inputs. In the example below output **X** implements the logic function (**B AND C**), while output **Y** implements (**A AND B AND C**). Can you see it?



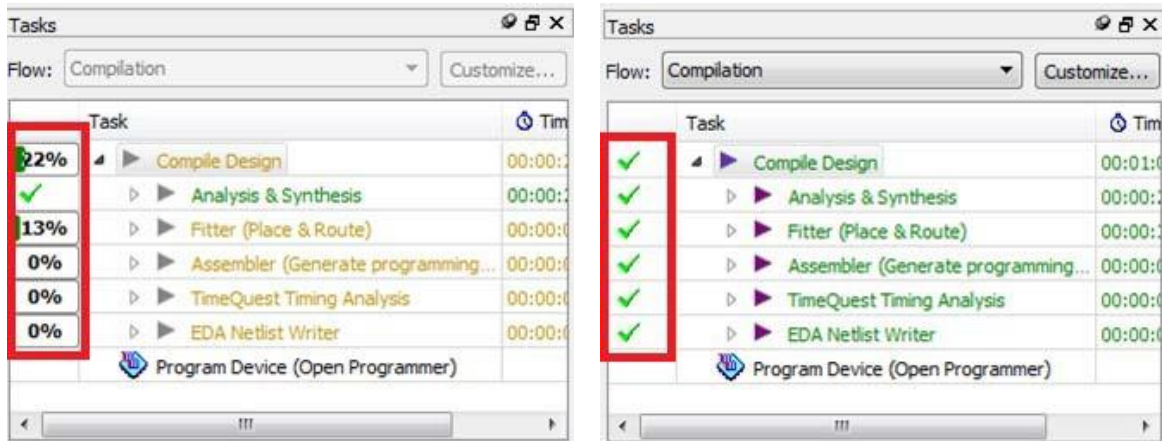
Compiling your Design

Once your design has been verified by simulation, the next step is to compile it. Compilation implies a series of processes the design has to go through. You may think of compilation as all the steps necessary to **translate** a schematic diagram into a file that can be downloaded to the reconfigurable fabric of the FPGA.

To compile your design double click on **Compile Design** in the flow panel on the left.

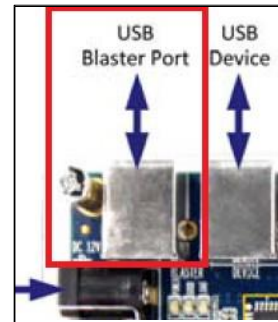


Observe how every task shows a percentage of progress as the process goes on. Unless your design has some errors, all tasks must show green ticks at the end of the compilation process.



Programming the FPGA

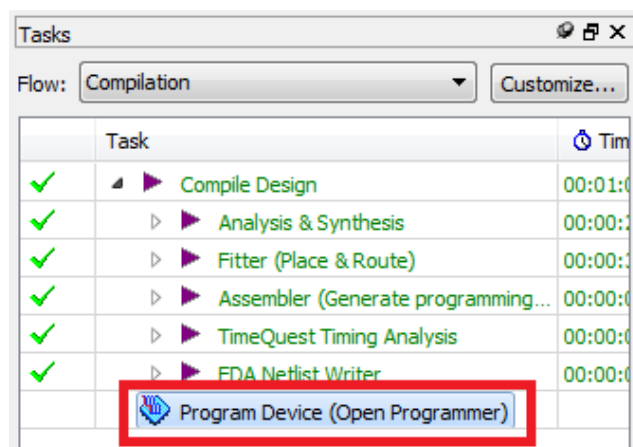
The FPGA device must be configured to implement the designed circuit. The required configuration file is generated at the end of the compilation process. The configuration data is transferred from the host computer (which runs the Quartus II software) to the board using the **USB Blaster Port** on the board. To use this connection, it is necessary to have the **USB-Blaster driver** installed. Before using the board, make sure that the USB cable is properly connected and turn on the power supply switch on the board.



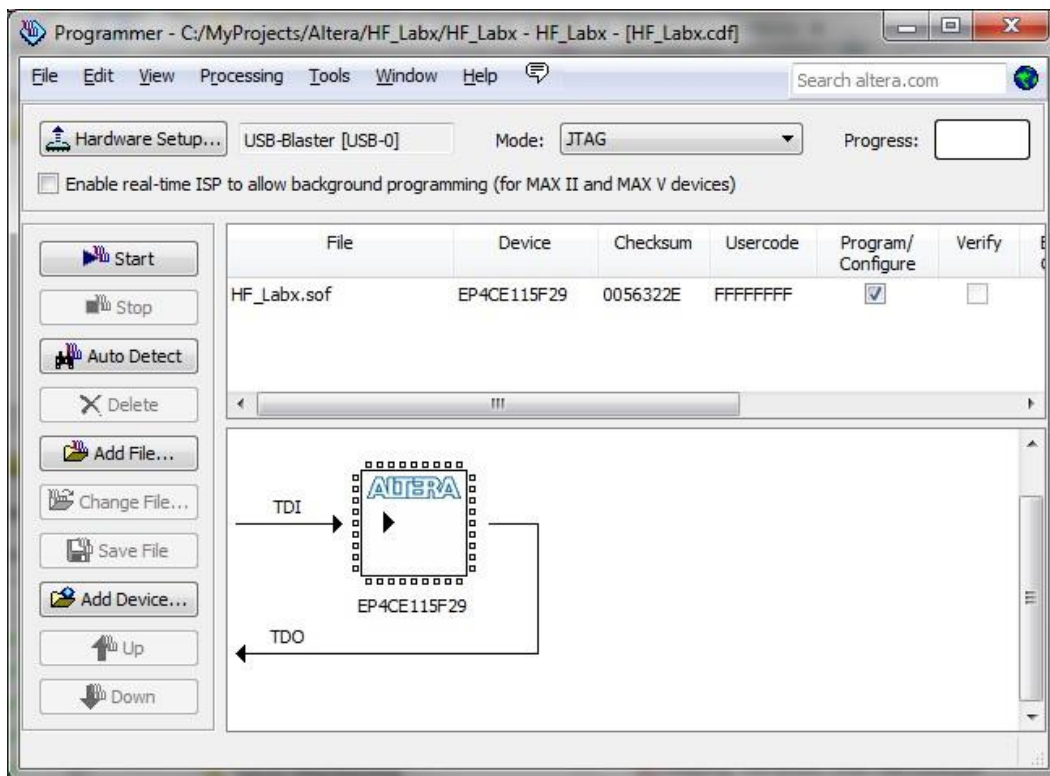
In JTAG mode, the configuration data is loaded directly into the FPGA device. If the FPGA is configured in this manner, it will retain its configuration as long as the power remains. The configuration information is lost when the power supply is turned off. Make sure switch **SW19** on the board is in the **RUN** position to select the JTAG mode.



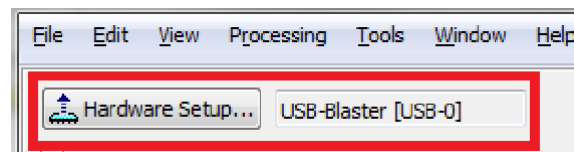
To invoke the programmer application, double click on **Program Device (Open Programmer)** in the Tasks panel.



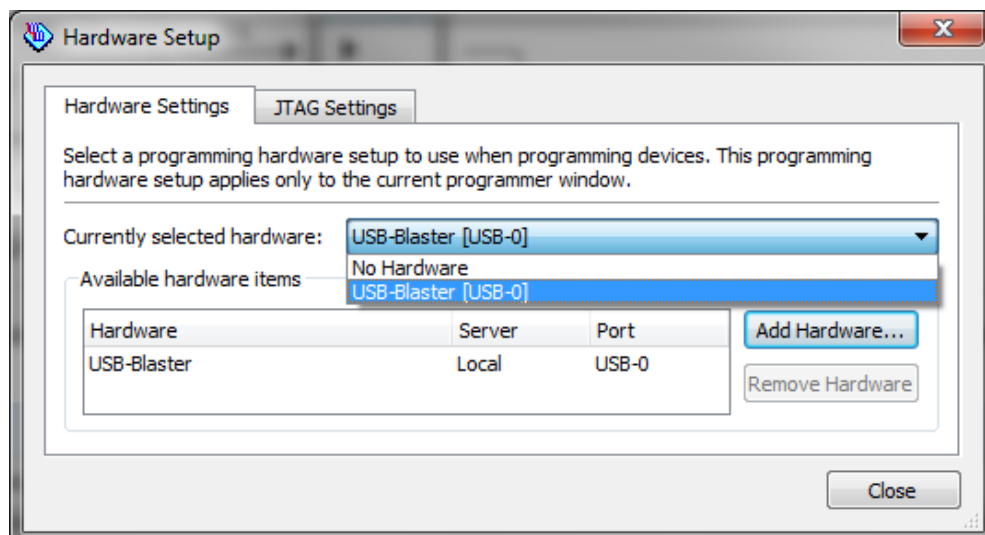
The programmer application allows you to download configuration files (.sof) to devices in a JTAG chain. Devices detected in the JTAG chain are shown in the bottom panel of the programmer window. In our case only one device is shown.



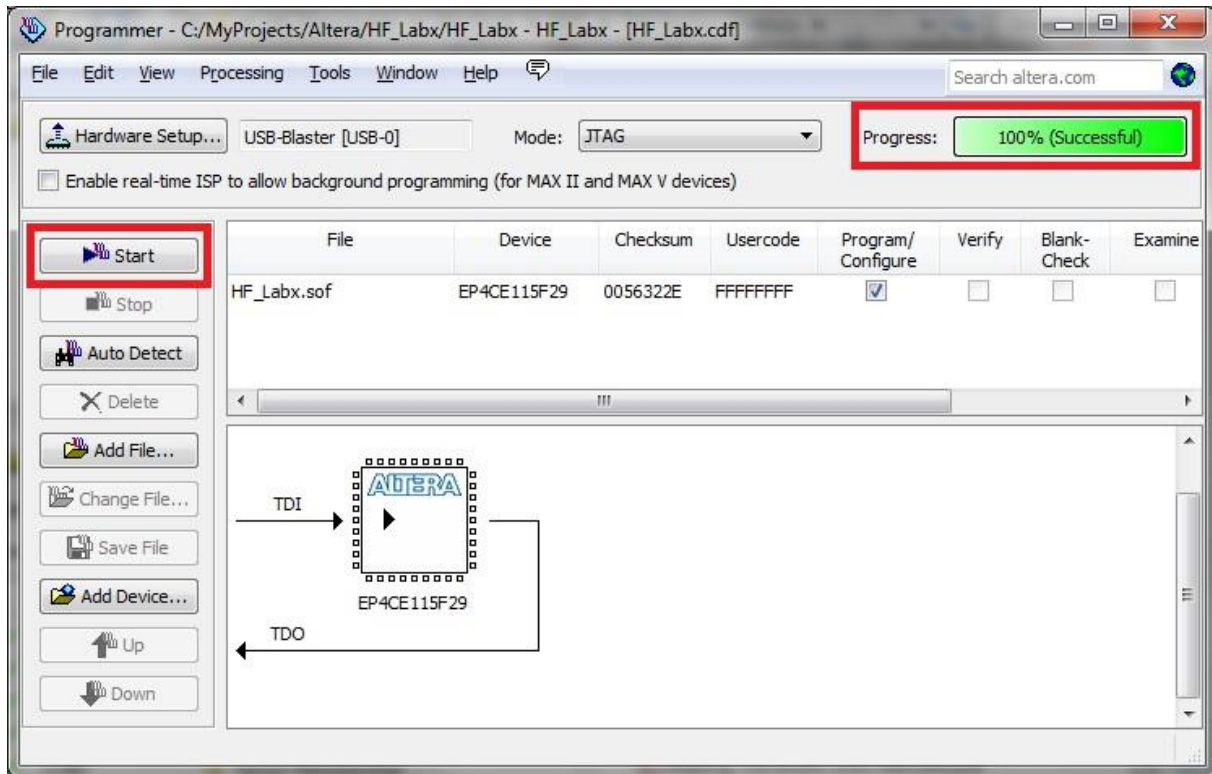
Make sure the system recognises the **USB-Blaster** port. It should appear in the **Hardware Setup** box.



If the port is not recognised, then press the **Hardware Setup** button, select **USB-Blaster** from the pull-down menu next to **Currently selected hardware** and press **Close**.



Back in the Programmer window press **Start** to download the configuration file to the FPGA. If the process is successful, then you should see a green, 100% progress status, and the FPGA should be implementing the function of your schematic diagram. You may now test your design by moving switches and observing the behaviour of the LEDs.



Afterword

Mastering a design tool is an ability that improves with practice. It is advisable that you spend as much time as possible using the QUARTUS II software. Use the Help menu when necessary and come to the lab with questions.